Supplementary Information

Discrimination transports, calibration does not: calibration drift in severity scoring across 187 intensive-care units

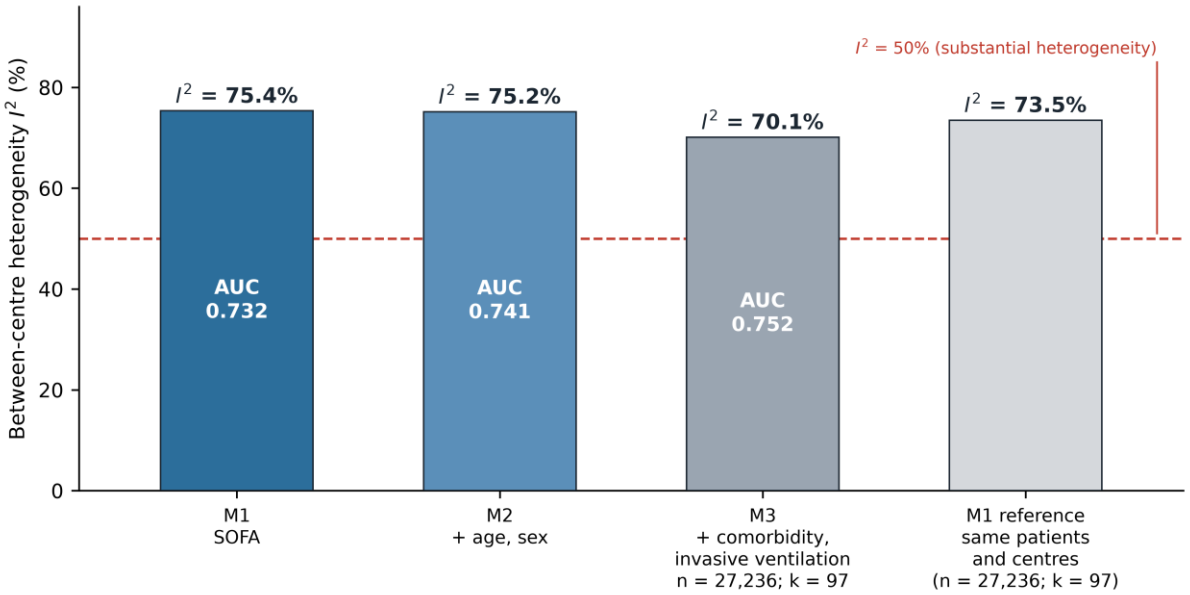
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Note on numbering. Supplementary figures and tables are numbered S1, S2 … separately from the main article. Supplementary Notes are not numbered, in accordance with the journal's supplementary material policy.

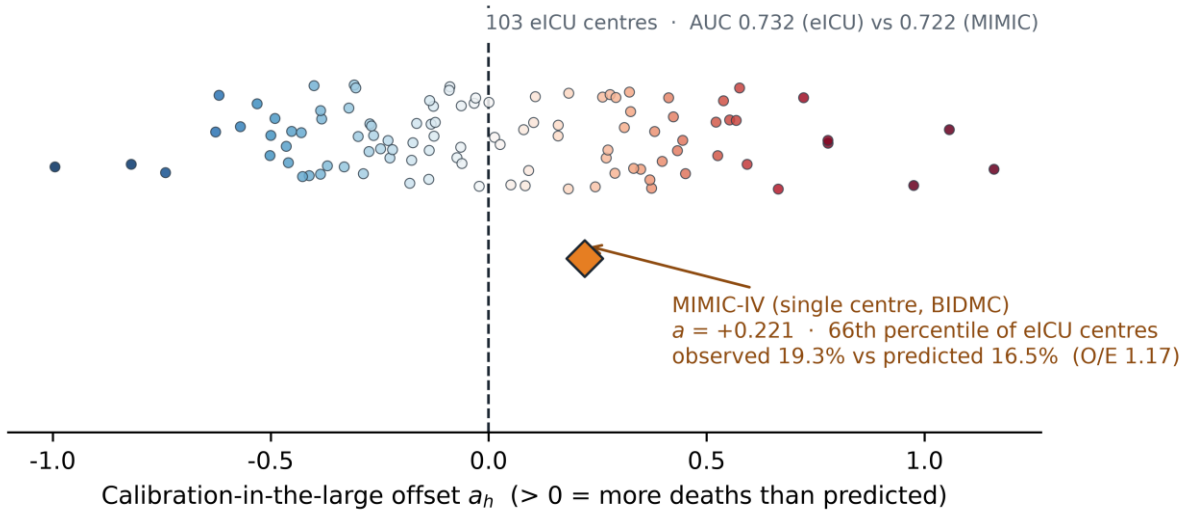
The analysis plans reproduced in the Supplementary Note were written before the corresponding analyses were executed. Their file timestamps, an explicit disclosure of one timestamp defect, and a post-hoc correction to the weighting of the shrinkage target are all included, so that the reader can verify what was specified before what was run.

Supplementary Figure S1. Case-mix ladder: between-centre heterogeneity across nested models.



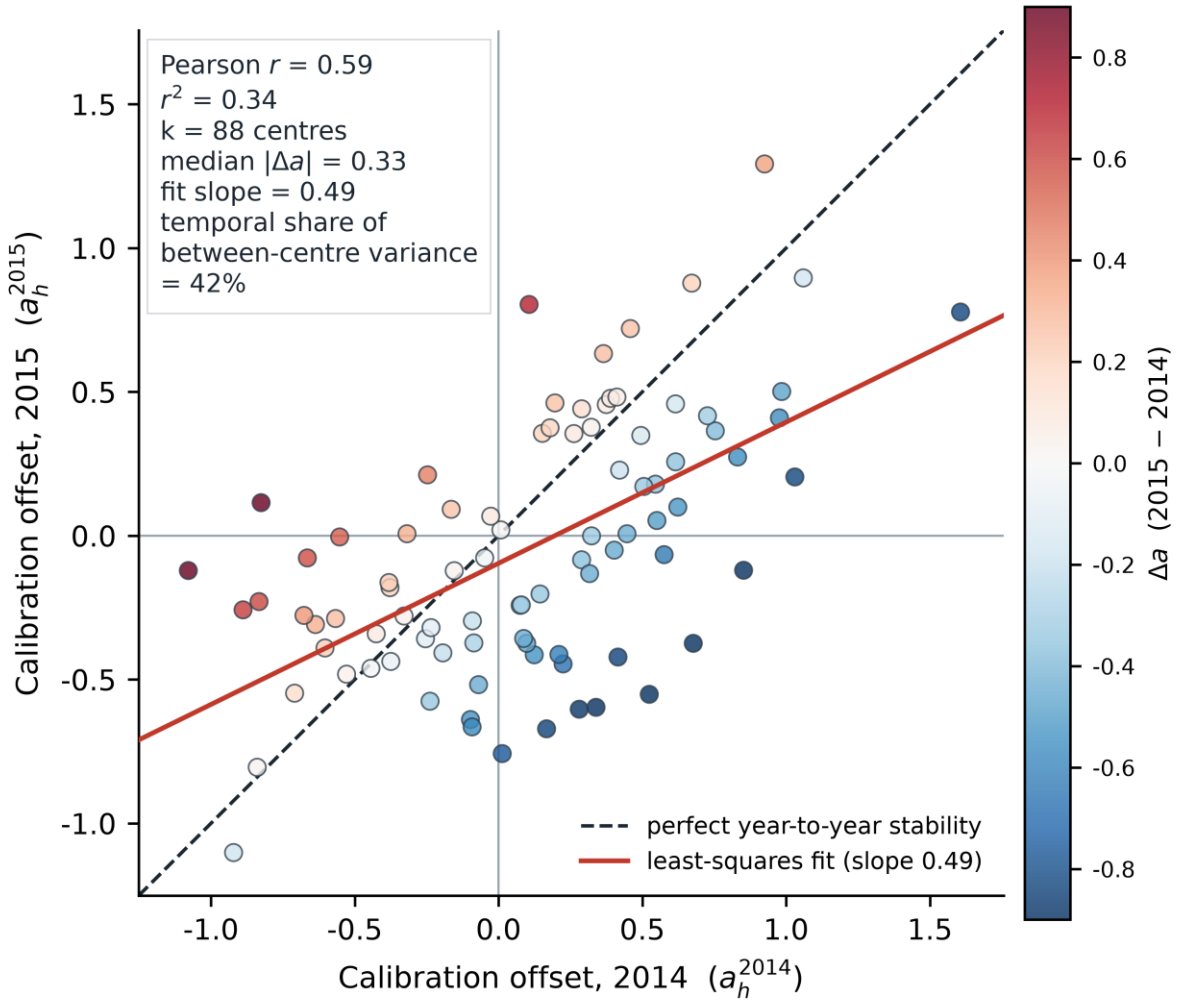
Supplementary Fig. S1. Measured case-mix barely reduces between-centre heterogeneity. Successive adjustment of the SOFA-based model: M1 = SOFA alone; M2 = + age, sex; M3 = + comorbidity burden (Charlson) + invasive ventilation. Bars show between-centre I^2 (random-effects meta-analysis, DerSimonian–Laird); numbers inside bars are the corresponding pooled AUCs. Adding age, sex, comorbidity burden and ventilation status raised the AUC by only **0.020** (0.732 → 0.752) while I^2 fell by just **5 percentage points** (75.4% → 70.1%). The red dashed line marks the conventional threshold for substantial heterogeneity (I^2 = 50%). **Note:** unmeasured case-mix (disease spectrum, referral patterns, socioeconomic status, pre-hospital course) cannot be excluded as a contributor. The remaining variation is therefore reported descriptively as **residual between-hospital variation** and is **NOT attributed to differences in quality of care**.

Supplementary Figure S2. Position of the single-centre MIMIC-IV cohort within the eICU-CRD distribution of calibration offsets.



Supplementary Fig. S2. Position of a single-centre development cohort within the multicentre distribution. The eICU-derived SOFA model (full-cohort coefficients) was applied to MIMIC-IV (8,579 patients, 1,654 deaths, 19.3%). MIMIC-IV is shown as the orange diamond; the grey points are the 103 eICU centres. MIMIC-IV had an AUC of 0.722 (0.732 within eICU), observed mortality 19.3% versus predicted 16.5% (O/E 1.17), and a calibration offset a of +0.221 — **the 66th percentile of the eICU distribution**. The single-centre cohort is therefore not an outlier; the systematic miscalibration commonly observed when transporting a single-centre model reflects ordinary between-centre drift rather than an anomalous development cohort.

Supplementary Figure S3. Year-to-year stability of the calibration offset within hospitals.



Supplementary Fig. S3. A centre's calibration offset is not a stable centre property.
eICU-CRD covers 2014–2015; **88 centres** contributed ≥50 patients and ≥5 deaths in each year. Each point is a centre: x = its calibration offset in 2014, y = its offset in 2015; colour = change (*Delta a* = 2015 − 2014). The dashed line is the identity line.
The two years correlated at **Pearson r = 0.586** (r^2 ≈ 34%; Spearman ρ = 0.563, P = 1.2×10^{-8}); the median within-centre |*Delta a*| was **0.331** (P90 0.831). Variance decomposition attributed **41.6%** of the total between-centre variance to the within-centre temporal component.
A centre's offset estimated in one year explains only about a third of its offset in the next — **any strategy relying on a one-off, fixed per-centre recalibration is inherently fragile**.

Supplementary Table S1. Cohort characteristics.

Characteristic	eICU-CRD (all)	eICU-CRD (103 centres)	MIMIC-IV
Patients, n	30,796	26,436	8,579
Hospitals, n	187	103	1
Male, n (%)	15,538 (50.5)	13,396 (50.7)	4,338 (50.6)
Age, years, mean (SD)	77.0 (7.7)	76.9 (7.7)	77.7 (8.3)
Charlson comorbidity index, mean (SD)	0.9 (1.3)	0.9 (1.3)	3.5 (2.5)
Missing data, n (%)			
└ Severity scores (SOFA / OASIS / SAPS-II)	0 (0)	0 (0)	0 (0)
└ Age, sex, invasive ventilation	0 (0)	0 (0)	0 (0)
└ In-hospital death (outcome)	0 (0)	0 (0)	0 (0)
└ Charlson comorbidity index	3,560 (11.6)	2,807 (10.6)	3 (0.04)
Invasive ventilation, n (%)	6,683 (21.7)	6,134 (23.2)	2,062 (24.0)
SOFA score, mean (SD)	4.3 (3.2)	4.4 (3.3)	5.5 (3.5)
OASIS score, median [IQR]	27 [22–34]	27 [22–35]	33 [28–39]
SAPS-II score, median [IQR]	35 [28–45]	35 [28–45]	41 [34–51]
ICU death, n (%)	2,325 (7.5)	2,094 (7.9)	1,014 (11.8)
In-hospital death, n (%)	3,901 (12.7)	3,486 (13.2)	1,654 (19.3)

Legend. Continuous variables are summarised as mean (SD) where the distribution was approximately symmetric in the full eICU-CRD cohort and as median [interquartile range] otherwise; the same summary is used for that row in all three columns. Percentages are column percentages and may not sum to 100 because of rounding. eICU-CRD (103 centres) is the subset of patients contributing to the centre-level analyses. **Missing data** are reported for every analysis variable; all variables except the Charlson comorbidity index are complete, so no imputation was performed. Comorbidity index enters only the descriptive case-mix analysis, which uses complete cases; every other analysis uses a set with no missing values.

Supplementary Table S2. Between-centre variance estimated under four frameworks.

Estimation framework	τ^2	I^2, %	Pooled mean (ā)	Relative difference from two-stage median
DerSimonian–Laird (two-stage)	0.1312	75.35	+0.030	—
Paule–Mandel (two-stage)	0.1261	75.35	+0.0304	−3.9%
REML (two-stage)	0.1299	75.35	+0.0301	−1.0%
One-stage multilevel logistic	0.0648	—	+0.0745	**−50.1%**

Legend. Two-stage estimates were obtained by random-effects meta-analysis of the 103 hospital-level calibration offsets. The one-stage estimate was obtained by fitting a random-intercept logistic regression to patient-level data (26,436 patients, 103 groups) by maximum *a posteriori* estimation; τ^2 is the square of the estimated random-effect standard deviation. Maximum *a posteriori* estimation of a variance component is sensitive to initialisation; repeated fits give standard deviations of 0.247–0.255 (τ^2 0.061–0.065), and an independent marginal-likelihood calculation gave $\tau^2 = 0.0706$. Pooled mean ā is reported with random-effects weights for the two-stage frameworks. **The pre-specified criterion** required the relative range across the three two-stage estimators to be < 25% (observed 3.9%, met) **and** the one-stage estimate to lie within 50% of their median (observed −50.1%, **not met**); the disagreement is reported as a negative finding.

Supplementary Table S3. Sensitivity of the empirical-Bayes result to the choice of between-centre variance.

Local patients (*m*)	$\tau^2 = 0.1312$ (DL)	$\tau^2 = 0.0706$ (marginal likelihood)	$\tau^2 = 0.0648$ (multilevel)
25	15.5% → **24.3%** (+9.1, P = 2.3×10^{-113})	15.5% → **21.9%** (+6.7, P = 2.7×10^{-64})	15.5% → **21.7%** (+6.5, P = 1.3×10^{-60})
50	19.7% → **26.4%** (+6.7, P = 1.4×10^{-61})	19.7% → **24.3%** (+4.6, P = 1.6×10^{-29})	19.7% → **23.8%** (+4.2, P = 1.4×10^{-24})
100	25.7% → **30.4%** (+4.7, P = 1.2×10^{-18})	25.7% → **27.7%** (+2.0, P = 4.6×10^{-4})	25.7% → **27.6%** (+1.9, P = 0.001)
200	32.3% → **34.4%** (+2.0, P = 0.035)	32.3% → **32.6%** (+0.2, P = 0.832)	32.3% → **32.5%** (+0.2, P = 0.889)

Legend. Each cell gives the success rate of naive local recalibration → that of empirical-Bayes shrinkage, with the difference in percentage points and the two-sided Wilcoxon signed-rank *P* value across paired (hospital, replicate) observations; $\alpha = 0.05$. Two hundred resampling attempts were used per cell, on the same draws for all three values of τ^2, so that the columns differ only in the variance used for shrinkage. The three candidate values were chosen to span the range produced by the estimation frameworks compared in Supplementary Table S2: 0.1312 (DerSimonian–Laird, used in all primary analyses), 0.0706 (marginal likelihood) and 0.0648 (one-stage multilevel). **The pre-specified criterion** — that shrinkage exceed naive recalibration at $m = 25$ — was met under all three values (all $P < 10^{-59}$). The magnitude of the advantage depends on τ^2 and the advantage is not statistically detectable at $m = 200$ for the two smaller values, which is why the strategy is described as one for small local samples rather than a general replacement for naive estimation.

Supplementary Note

Prespecified analysis plans: index, chronology and disclosures

Manuscript: *Discrimination transports, calibration does not: calibration drift in severity scoring across 187 intensive-care units*

Prepared 2026-10-19. This supplement contains, verbatim, the analysis plans that were written **before** the corresponding analyses were executed, together with file-system timestamps, an explicit disclosure of one timestamp defect, and an English summary of each plan.

> **Note on language.** The original plans are reproduced **in their original language (Chinese), unaltered**, because they are the primary record of what was specified before what was computed. Translating them at this stage would break the chain between plan and result. An English summary of each plan is provided below so that the content is accessible to readers of the submitted manuscript.

S1.1 Chronology

#	Plan file	Written	Result file	Result written	Interval
P0	路B_预注册.md (feasibility probe)	2026-10-19 06:25:26	probe output (I^2 = 75.2%, 112 centres)	same run	—
P1	分析计划_预注册.md (analyses A–D)	2026-10-19 06:34:11	drift_results.json	06:34:27	**16 s**
P2	学习曲线_预注册.md (analysis E)	*see S1.2*	learning_curve_raw.csv	07:06:05	see S1.2
P3	分析计划_第二批.md (analyses F–H)	2026-10-19 07:38:50	batch2_results.json	07:38:53	**3 s**
P4	分析计划_分析I.md (analysis I)	2026-10-19 07:56:49	shrinkage_raw.csv	08:12:05	15 min
P5	分析计划_分析J.md (analysis J)	2026-10-19 09:41:03	mimic_external_summary.json	09:41:43	**40 s**
P6	分析计划_分析K.md (analysis K)	2026-10-19 10:43:49	tau2_framework.json	10:43:54	**5 s**

Six of the seven plans carry file-system timestamps that are demonstrably earlier than the outputs they govern, several by seconds.

S1.2 Disclosure of a timestamp defect (plan P2)

We disclose this rather than conceal it, because it is exactly the kind of defect this supplement exists to surface.

What happened. The content of 学习曲线_预注册.md is generated programmatically from the module docstring of analysis_learning_curve.py. That docstring has **never been edited**. However, a caching feature was later added to the script for development speed, and the script **rewrites the plan file on every run**. This refreshed the file's modification time to **2026-10-19 11:16:14**, which is *later* than the output it governs (learning_curve_raw.csv, 07:06:05). **The modification time of P2 is therefore no longer admissible as evidence of precedence.**

What still supports precedence. The run that first produced learning_curve_raw.csv at 07:06:05 executed, by program order, the write of the plan file *before* entering any computation. The plan therefore existed, in the content reproduced in this supplement, no later than 07:06:05.

Remediation. (i) A provenance note was appended to the plan file itself, recording the defect and the supporting argument. (ii) The writing of every plan file was changed to be **idempotent** — an existing plan is never overwritten — so that file timestamps can no longer be disturbed by re-runs. This was applied to all seven analysis scripts.

S1.3 Classification of analyses: exploratory versus confirmatory

This classification was fixed after the analyses were complete and is reported for transparency. It is **not** a claim that the plans in S1.1 were all confirmatory.

Class	Analyses	Basis	Outcome
Exploratory / descriptive	A, B, C, D, F, G, H	No pre-specified threshold; magnitudes estimated and described	Results reported as descriptive estimates with no hypothesis test

Confirmatory (pre-specified criterion)	E, I, J, K	Each plan stated an explicit criterion *and* an explicit instruction for the case where it was not met	E: **criterion not met** (success rate never reached 50%) — reported as such. I, J: criterion met. K: **criterion not met** (two-stage and one-stage τ^2 differ by ~2-fold) — reported as such

Two of the four pre-specified criteria were not met, and both are reported in the manuscript as negative findings rather than being reframed post hoc. In particular:

- Analysis E pre-specified "report the smallest m at which the success rate reaches 50%; if none is reached, state so." The answer was *no such m exists within 200 local patients*, which is reported.
- Analysis K pre-specified agreement between two-stage and one-stage estimators of τ^2 (relative range < 25%, and within 50% of the meta-analytic median). This was **not** achieved (the multilevel estimate was 50.1% lower than the two-stage median), and the disagreement is reported together with a proposed mechanism and a sensitivity analysis of the main conclusion.

The primary descriptive findings (heterogeneity magnitude, its relation to measured case-mix, its temporal instability and its clinical consequences) are **estimates, not tests**, and are presented as such.

S1.4 English summaries of the plans

P0 — Feasibility probe (路B_预注册.md)

Question: does the calibration of an acute-severity score vary across individual centres? *Design:* pooled logistic model (SOFA + age + sex) fitted on the full eICU-CRD cohort; per-centre calibration-in-the-large offset a_h; between-centre heterogeneity by random-effects meta-analysis. *Pre-specified criterion:* **I^2 > 50% AND ≥ 100 centres with ≥ 100 patients** → the research direction is viable. *Outcome: met* (I^2 = 75.2%, 112 centres).

P1 — Analyses A–D (分析计划_预注册.md)

A–B: per-centre drift quantified for SOFA, OASIS and SAPS-II, including per-centre calibration slope. *C:* a case-mix ladder (M1 = score; M2 = + age, sex; M3 = + comorbidity, invasive ventilation) testing whether I^2 falls as covariates are added. *D:* positioning of the single-centre MIMIC-IV cohort within the eICU distribution. *Wording red line:* variation not explained by measured case-mix must be termed **residual between-hospital variation** and must **not** be attributed to quality of care.

P2 — Analysis E, learning curve (学习曲线_预注册.md)

For each centre (n ≥ 200) and each m ∈ {0, 25, 50, 100, 200}, requiring n ≥ 2m, m patients were drawn for local recalibration and the remainder used for evaluation; each (centre, m) cell repeated 100 times. Two strategies compared: intercept-only, and intercept plus calibration slope. *Pre-specified criterion:* report the smallest m at which the success rate reaches 50%; if none is reached, state so. *Outcome:* **not met** — the success rate never reached 50% within 200 patients.

P3 — Analyses F–H (分析计划_第二批.md)

F: split-half temporal stability — per-centre offset in 2014 versus 2015 (centres contributing ≥ 50 patients and ≥ 5 deaths in each year), reported as correlation, median |Δa| and the temporal share of between-centre variance. *G:* stratification of heterogeneity by centre size tertiles. *H:* clinical impact — per-centre net benefit and positive predictive value at decision thresholds 0.10–0.40. *Wording red line:* a low split-half correlation indicates that the apparent between-centre variation contains an unstable component; it must **not** be described as unstable quality of care.

P4 — Analysis I, shrinkage (分析计划_分析I.md)

Compares four strategies — no recalibration (S0), naive intercept-only (S1), naive intercept plus slope (S2), and **empirical-Bayes shrinkage** (S3), $a_{EB} = \bar{a} + \lambda(a_{MLE} - \bar{a})$ with $\lambda = \tau^2/(\tau^2 + SE^2)$ — with $\bar{a}$ and τ^2 estimated by **leave-one-centre-out** cross-fitting so that no centre's evaluation data enters its own shrinkage parameters. *Pre-specified criterion:* at m = 25 the ±10% success rate of S3 must exceed S1, and at m ≥ 100 S3 must not be worse than S1 by more than 5 percentage points. *Outcome: met.*

P5 — Analysis J, cross-database transfer of the shrinkage parameters (分析计划_分析J.md)

MIMIC-IV (8,579 patients) treated as a **single new centre**; the shrinkage target $\bar{a} = +0.030$ and $\tau^2 = 0.131$ were **fixed from eICU and never re-estimated on MIMIC-IV**; the true offset to recover was a *= +0.221.* Pre-specified criterion: *at m = 25 the ±10% success rate of S3 must exceed both S1 and S0.* Outcome: met. Wording red line: *MIMIC-IV is a single-centre database; this analysis tests transfer of the shrinkage parameters across databases and is not* multicentre external validation of the strategy.*

P6 — Analysis K, τ^2 estimation framework (分析计划_分析K.md)

The between-centre variance τ^2 re-estimated under four frameworks — DerSimonian-Laird, Paule-Mandel, REML (two-stage random-effects meta-analysis) and a one-stage random-intercept multilevel logistic model fitted on patient-level data. *Pre-specified criterion:* relative range of τ^2 across the three two-stage estimators < 25%, and the multilevel estimate within 50% of their median. *Outcome:* first clause **met** (range 3.9%); second clause **not met** (multilevel τ^2 = 0.065, i.e. 50.1% lower). Reported as a disagreement, with a proposed mechanism (the fixed-slope two-stage estimator absorbing part of the between-centre slope variation) and a sensitivity analysis showing that the main conclusion holds under all three τ^2 values.

S1.5 Why no prospective registry registration is claimed

This study is a **retrospective secondary analysis of already-public databases**. All analyses were executed before this supplement was assembled, so no prospective registry entry (OSF or otherwise) can honestly be claimed, and none is. The plans reproduced here constitute the authors' contemporaneous record of analysis specification. For transparency we note that a registry entry made at this stage would be a *post-hoc* registration and would carry none of the protections that prospective registration provides.

Two criteria pre-specified in this supplement (P2 and P6) were **not met**, and are reported as negative findings in the manuscript.

S1.6 Disclosure: a post-hoc correction to the weighting of the shrinkage target

This is disclosed rather than quietly applied, and it affects the numeric values of analyses I and J.

What was wrong. The empirical-Bayes shrinkage estimator is written in the plans as `a_EB = ā + λ(a_MLE − ā)` with `λ = τ²/(τ² + SE²)`. The shrinkage intensity λ came from the hierarchical model `a_h ~ N(ā, τ² + SE²)`, but the shrinkage target ā had been estimated with **fixed-effect** weights (`1/SE²`), which is the estimate implied by `τ² = 0`. The two halves of the estimator therefore rested on mutually contradictory assumptions.

Why the corrected value is the right one. Under the hierarchical model that λ assumes, the maximum likelihood estimate of ā is the **random-effects weighted** mean `Σ(a_h/(SE²+τ²)) / Σ(1/(SE²+τ²))`. The same programme already used random-effects weights for the pooled standard error `seRE` and for the pooled mean reported in analysis K, so the fixed-effect value was additionally internally inconsistent with quantities reported elsewhere in the same manuscript.

Does this violate the pre-specified plans? The plans specify *what* ā and τ^2 are ("estimated leave-one-centre-out" for analysis I; "fixed from eICU, never re-estimated on MIMIC-IV" for analysis J) but **they do not specify the weighting scheme used to compute ā**. No pre-specified criterion is changed by this correction, and no threshold is moved. The correction resolves an ambiguity in the plans; it is not a deviation from them.

Effect on the results.

Analysis J (MIMIC-IV), S3 success rate (±10%)	ā fixed-effect (original)	ā random-effects (corrected)
m = 25	53.5%	**47.0%**
m = 50	49.0%	46.0%
m = 100	55.0%	56.0%
m = 200	68.5%	66.0%
m = 400	75.0%	75.0%

The pre-specified criterion for analysis J (at m = 25, S3 must exceed both S1 and S0) is **met under both values** and is therefore not affected. The paired advantage over naive recalibration at m = 25 falls from +30.0 to +23.5 percentage

points ($P = 6.3 \times 10^{-7}$). Because the corrected success rate at m = 25 now sits below naive recalibration at m = 200, the comparative statement about the reduction in local sample size is reported from the corrected values throughout.

A pre-existing inconsistency is reported for transparency. Under the original fixed-effect value the analyses all reported smaller standard errors of the pooled estimate than were actually used; this did not affect any point estimate reported outside analyses I and J, and no descriptive result in analyses A–H or K changes.

Not changed. All descriptive results (A–H and K) are unaffected: they depend on τ^2 and I^2, whose estimators were already using the correct fixed-effect weights for Q. Re-running analysis A–D after the correction reproduced **every pre-existing field exactly**, with the random-effects mean added as a new field.

S1.7 Specification details left open in the plans

These are recorded for completeness. Neither carries any degrees of freedom that could have been resolved in a direction favourable to the findings, and no pre-specified threshold is affected.

#	Left unspecified in the plan	What was done	Why this is not consequential
1	**Number of resamples in analysis I.** The plan for analysis I fixes the four strategies, the leave-one-centre-out design, the metrics and the criterion, but never states how many times each (centre, *m*) cell is resampled. (Analysis J's plan does state it explicitly — 200 — and the learning-curve plan states 100; both were followed.)	**200 resamples per cell**	More resamples reduce Monte-Carlo error in every arm equally. A different count would change the precision of the estimate, not its direction, and cannot move a strategy relative to the pre-specified threshold.
2	**Which paired test.** The plans for analyses I and J specify a *paired, two-sided* comparison of success rates without naming the test.	**Wilcoxon signed-rank test, two-sided**	Selected as the standard paired non-parametric test and stated as such in Methods. A paired *t*-test on a 19,678-pair difference that is plainly non-normal would be the less appropriate choice; neither test can reverse a difference of the size observed (e.g. +9.05 percentage points).

The wording of both plans, unaltered, is reproduced in this item so that a reader can verify what was and was not specified.